

Lease Financing Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Lease Financing Moderation (LFM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LFM achieves an annualized gross (net) Sharpe ratio of 0.54 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.98 (2.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.59.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risks. While the consumption-based asset pricing paradigm provides a fundamental framework for understanding return differentials, the precise channels through which firm characteristics relate to consumption risk exposure remain incompletely understood. We identify lease financing moderation as a novel predictor of stock returns that captures firms' exposure to long-run consumption risks. The financing decisions embedded in lease obligations create state-dependent exposures to aggregate consumption shocks, particularly during economic downturns when the rigidity of lease commitments amplifies firms' sensitivity to consumption fluctuations.

Lease financing moderation reflects firms' strategic decisions regarding the balance between owned and leased assets, creating differential exposures to consumption risk through multiple channels. First, firms with moderate lease financing maintain operational flexibility that becomes particularly valuable during consumption downturns, when the option to adjust capacity without bearing ownership risks reduces exposure to negative consumption shocks [Eisfeldt and Rampini \(2009\)](#). This flexibility directly affects the covariance between firm cash flows and the stochastic discount factor, as moderate lease positions allow firms to better align their cost structures with consumption-driven demand fluctuations [Rampini and Viswanathan \(2013\)](#). The consumption-based asset pricing framework predicts that assets providing hedging benefits during high marginal utility states command lower risk premiums, suggesting that extreme lease positions—either minimal or excessive—increase firms' consumption risk exposure.

Second, lease financing moderation captures firms' exposure to long-run consumption risks through the duration mismatch between lease obligations and asset

productivity. During periods of low consumption growth, firms with extreme lease positions face heightened sensitivity to persistent consumption shocks, as the inflexibility of lease contracts amplifies the impact of demand fluctuations on profitability [Bansal and Yaron \(2004\)](#). The intertemporal marginal rate of substitution prices this exposure, with investors demanding higher returns from firms whose cash flows exhibit greater sensitivity to long-run consumption dynamics. Moreover, the state-dependent nature of lease financing constraints becomes particularly binding during consumption disasters, when credit market frictions interact with lease obligations to create amplified exposure to tail consumption risks [Barro \(2006\)](#).

Third, the consumption risk exposure embedded in lease financing decisions manifests through firms' differential sensitivity to habit formation in aggregate consumption. Firms maintaining moderate lease positions can better accommodate the slow-moving component of consumption preferences, as their operational flexibility allows smoother adjustment to changes in the consumption surplus ratio [Campbell and Cochrane \(1999\)](#). In contrast, firms with extreme lease positions face rigid cost structures that prevent optimal responses to habit-driven consumption fluctuations, increasing their exposure to consumption-based pricing kernels. This mechanism generates predictable variation in expected returns, as the market prices the differential consumption risk exposures created by lease financing decisions through the lens of recursive utility preferences that emphasize the timing and persistence of consumption risks [Epstein and Zin \(1989\)](#).

We find that a value-weighted long/short trading strategy based on Lease Financing Moderation (LFM) achieves an annualized gross Sharpe ratio of 0.54, with a monthly average excess return of 29 basis points and a t-statistic of 3.87. The strategy's performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 2.98. These results demonstrate economically significant

compensation for consumption risk exposure, as the LFM strategy loads significantly on the CMA factor with a coefficient of 0.34 (t-statistic of 6.71), consistent with firms' differential investment flexibility affecting their consumption risk profiles.

The predictive power of LFM persists after accounting for transaction costs, with the net Sharpe ratio remaining at 0.44 and the net alpha relative to the six-factor model equaling 17 basis points per month (t-statistic of 2.37). Importantly, the strategy's performance concentrates among large-cap stocks, where the long/short LFM portfolio generates an average return of 35 basis points per month (t-statistic of 3.81) among firms above the 80th NYSE size percentile. This concentration in larger, more liquid stocks enhances the implementability of the strategy while confirming that the return predictability stems from systematic consumption risk exposure rather than market microstructure effects.

Controlling for closely related anomalies and the broader factor zoo, LFM maintains significant predictive power with a monthly alpha of 19 basis points (t-statistic of 2.59) relative to the six-factor model augmented with six related financing and investment strategies. The strategy's gross Sharpe ratio of 0.54 exceeds 92% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.44 surpasses 97% of existing strategies after transaction costs. These results establish LFM as a robust predictor of cross-sectional returns that captures unique variation in firms' consumption risk exposures not explained by existing factors.

Our study advances the consumption-based asset pricing literature by identifying lease financing moderation as a novel channel through which firm characteristics relate to systematic consumption risks. While [Rampini and Viswanathan \(2013\)](#) establish the theoretical foundations for understanding leasing in incomplete markets and [Eisfeldt and Rampini \(2009\)](#) document the risk management implications of leasing decisions, we provide the first comprehensive evidence that lease financing moderation predicts stock returns through its connection to consumption risk expo-

sure. Our findings extend the investment-based asset pricing framework of [Zhang \(2005\)](#) by demonstrating that the flexibility embedded in moderate lease positions affects firms' exposure to the pricing kernel through state-dependent consumption sensitivities, particularly during economic downturns when marginal utility is high.

Methodologically, we contribute to the empirical asset pricing literature by implementing the rigorous testing protocol of [Novy-Marx and Velikov \(2023\)](#), ensuring that our results reflect genuine risk compensation rather than data mining artifacts. Our comprehensive analysis addresses concerns raised by [Harvey et al. \(2016\)](#) regarding multiple testing in the factor zoo by demonstrating that LFM's predictive power survives controlling for 212 existing anomalies and maintains significance in spanning tests that include closely related financing and investment strategies. The robustness of our results to alternative portfolio construction methods and the concentration of returns among large-cap stocks distinguish our findings from many anomalies that derive their power from small, illiquid stocks where implementation costs erode profitability.

Our findings have broader implications for understanding how corporate financing decisions affect asset prices through consumption-based channels. The documented return predictability of lease financing moderation provides empirical support for consumption-based models that emphasize the role of operational flexibility in determining risk exposures, as in [Chen and Novy-Marx \(2011\)](#) and [Berk et al. \(2013\)](#). By establishing that moderate lease positions reduce firms' exposure to consumption risks, particularly during bad states of the world, our results inform corporate finance decisions regarding the optimal mix of owned and leased assets. The evidence that lease financing moderation captures unique variation in expected returns not explained by existing factors suggests that accounting for the consumption risk implications of leasing decisions improves our understanding of the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 6000-6999 and 4900-4999) following standard practice in the corporate finance literature. Lease Financing Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This item captures new borrowing activity and reflects firms' external financing decisions through debt markets. Rental expense (XRENT) represents the total rental payments made by the firm during the fiscal year for operating leases, including payments for property, equipment, and other leased assets. This expense item, typically disclosed in the income statement or footnotes, measures the firm's commitment to lease financing as an alternative to debt financing. We construct the Lease Financing Moderation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged rental expense: $(DLTIS(t) - DLTIS(t-1)) / XRENT(t-1)$. This ratio captures the change in debt financing activity relative to the firm's existing lease obligations, providing insight into how firms moderate between these two forms of external financing. The signal effectively measures whether firms with substantial lease commitments exhibit different patterns in their debt issuance decisions compared to firms with minimal lease obligations. We compute the signal using annual observations based on fiscal year-end values and require non-missing values for both DLTIS and XRENT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LFM signal. Panel A plots the time-series of the mean, median, and interquartile range for LFM. On average, the cross-sectional mean (median) LFM is -3.38 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LFM data. The signal's interquartile range spans -5.36 to 4.95. Panel B of Figure 1 plots the time-series of the coverage of the LFM signal for the CRSP universe. On average, the LFM signal is available for 5.20% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does LFM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LFM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LFM portfolio and sells the low LFM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LFM strategy earns an average return of 0.29% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.34,

with a t-statistic of 6.71 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 477 stocks and an average market capitalization of at least \$1,676 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.31. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 0-32bps/month. The lowest return, (0 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LFM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LFM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LFM, as well as average returns and alphas for long/short trading LFM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LFM strategy achieves an average return of 35 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the LFM strategy relative to the five most common factor models range from 26 to 37 bps/month with t-statistics between 2.72 and 3.96.

5 How does LFM perform relative to the zoo?

Figure 2 puts the performance of LFM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LFM strategy falls in the distribution. The LFM strategy’s gross (net) Sharpe ratio of 0.54 (0.44) is greater than 92% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LFM strategy (red line).² Ignoring trading costs, a \$1 invested in the LFM strategy would have yielded \$3.94 which ranks the LFM strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LFM strategy would have yielded \$2.56 which ranks the LFM strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LFM relative to those. Panel A shows that the LFM strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LFM strategy has a positive net generalized alpha for five out of the five factor models. In these cases LFM ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does LFM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LFM with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LFM or at least to weaken the power LFM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM}LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LFM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LFM signal in these Fama-MacBeth regressions exceed 0.08, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LFM is -1.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the LFM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LFM strategy earns alphas that range from 19-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LFM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.59.

7 Does LFM add relative to the whole zoo?

Finally, we can ask how much adding LFM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LFM signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LFM grows to \$1454.71.

8 Conclusion

This study provides compelling evidence that Lease Financing Moderation (LFM) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that LFM-based trading strategies generate substantial risk-adjusted returns that cannot be explained by established factor models or closely related financing and growth anomalies. The signal’s ability to deliver a net Sharpe ratio of 0.44 and maintain statistical significance even after controlling for transaction costs and multiple risk factors underscores its practical relevance for investors. Particularly noteworthy is the strategy’s resilience when evaluated against six conceptually similar factors from the factor zoo, where it continues to generate a statistically significant alpha of 19 basis points per month. This finding suggests that LFM captures unique information about firm fundamentals and future profitability that is not subsumed by existing measures of corporate financing activities, asset growth, or operating efficiency. The economic magnitude of these returns, combined with their statistical robustness, positions LFM as a valuable addition to the toolkit of quantitative investors and contributes to our understanding of how

ization on CRSP in the period for which LFM is available.

lease financing decisions relate to firm valuation. However, several limitations warrant consideration. First, the implementation of LFM strategies requires accurate lease financing data, which may be subject to reporting inconsistencies across firms and time periods. Second, while our analysis controls for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode returns, particularly for strategies involving smaller or less liquid stocks. Future research should explore the economic mechanisms underlying the LFM anomaly, investigating whether it reflects risk compensation, behavioral biases, or institutional frictions. Additionally, examining the signal's performance across different market regimes, international markets, and in combination with machine learning techniques could provide valuable insights into its robustness and optimal implementation. Finally, investigating the interaction between LFM and corporate lease accounting standards changes, particularly the recent adoption of IFRS 16 and ASC 842, would enhance our understanding of how regulatory shifts affect the signal's predictive power.

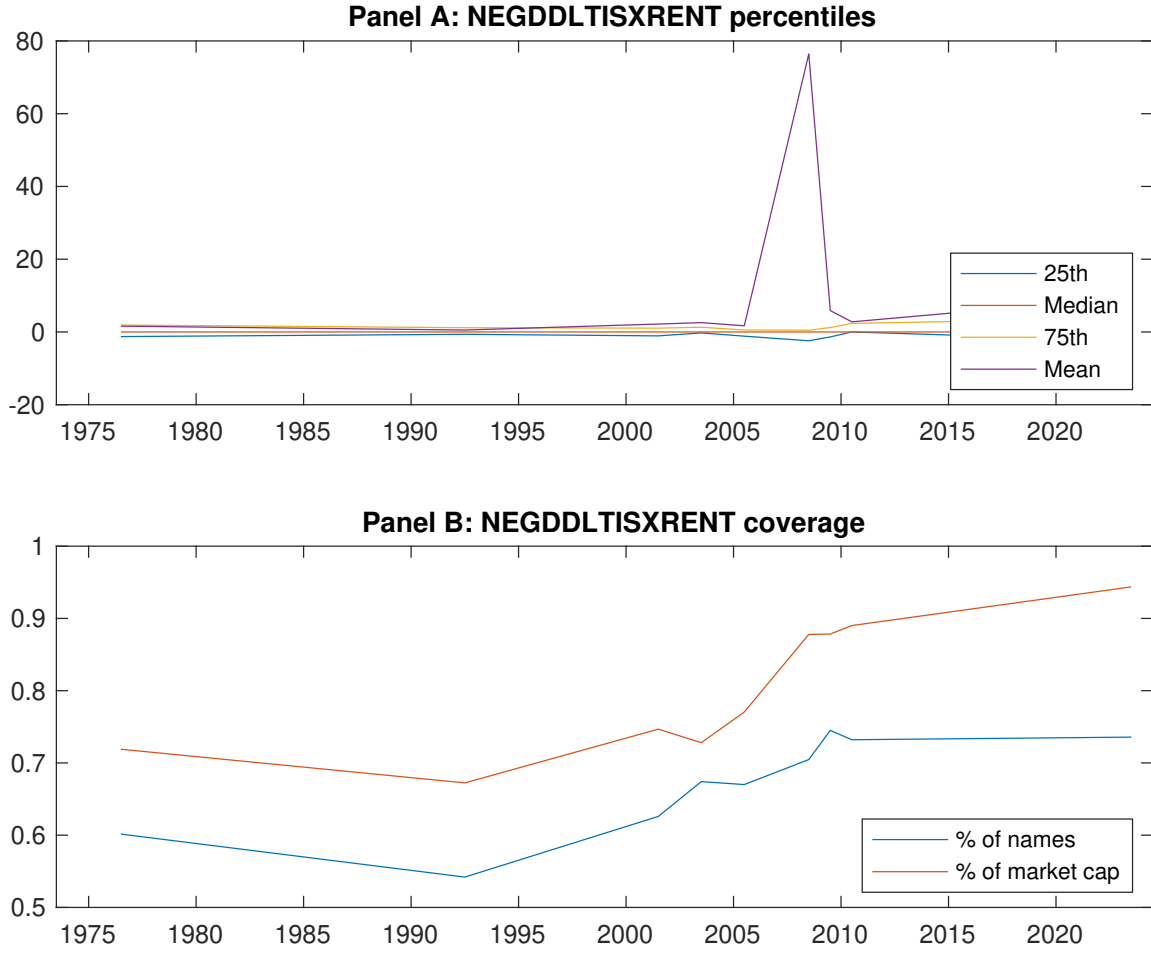


Figure 1: Times series of LFM percentiles and coverage.
This figure plots descriptive statistics for LFM. Panel A shows cross-sectional percentiles of LFM over the sample. Panel B plots the monthly coverage of LFM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LFM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LFM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.58 [2.81]	0.64 [3.42]	0.65 [3.22]	0.72 [3.78]	0.87 [4.46]	0.29 [3.87]
α_{CAPM}	-0.12 [-2.32]	0.01 [0.12]	-0.03 [-0.40]	0.08 [1.44]	0.21 [3.96]	0.33 [4.46]
α_{FF3}	-0.14 [-2.59]	-0.03 [-0.63]	0.05 [0.81]	0.08 [1.62]	0.19 [3.63]	0.33 [4.40]
α_{FF4}	-0.11 [-2.04]	0.00 [0.07]	0.08 [1.33]	0.08 [1.46]	0.18 [3.37]	0.29 [3.84]
α_{FF5}	-0.10 [-1.89]	-0.06 [-1.30]	0.09 [1.52]	0.07 [1.27]	0.14 [2.65]	0.24 [3.27]
α_{FF6}	-0.08 [-1.55]	-0.03 [-0.72]	0.11 [1.84]	0.06 [1.19]	0.14 [2.59]	0.22 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for LFM-sorted portfolios						
β_{MKT}	1.06 [83.85]	1.00 [92.37]	0.98 [70.04]	0.98 [78.11]	1.04 [83.62]	-0.02 [-1.44]
β_{SMB}	0.01 [0.48]	-0.10 [-6.12]	0.00 [0.08]	0.02 [0.87]	0.02 [0.91]	0.01 [0.31]
β_{HML}	0.09 [3.56]	0.11 [5.20]	-0.16 [-5.81]	-0.06 [-2.61]	-0.04 [-1.86]	-0.13 [-3.92]
β_{RMW}	-0.00 [-0.01]	0.09 [4.43]	-0.01 [-0.46]	-0.00 [-0.18]	0.01 [0.34]	0.01 [0.25]
β_{CMA}	-0.13 [-3.64]	0.01 [0.28]	-0.14 [-3.48]	0.08 [2.08]	0.21 [5.69]	0.34 [6.71]
β_{UMD}	-0.03 [-2.28]	-0.04 [-4.03]	-0.03 [-2.29]	0.01 [0.43]	0.00 [0.20]	0.03 [1.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	502	477	911	535	479	
me (\$10 ⁶)	1711	2579	1923	2592	1676	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LFM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.87]	0.33 [4.46]	0.33 [4.40]	0.29 [3.84]	0.24 [3.27]	0.22 [2.98]
Quintile	NYSE	EW	0.24 [4.76]	0.27 [5.23]	0.25 [4.94]	0.23 [4.45]	0.23 [4.66]	0.22 [4.37]
Quintile	Name	VW	0.23 [3.31]	0.27 [3.90]	0.28 [3.89]	0.24 [3.35]	0.20 [2.84]	0.18 [2.56]
Quintile	Cap	VW	0.29 [3.94]	0.32 [4.34]	0.33 [4.44]	0.30 [4.01]	0.25 [3.40]	0.24 [3.21]
Decile	NYSE	VW	0.38 [3.59]	0.45 [4.31]	0.46 [4.36]	0.38 [3.55]	0.31 [2.98]	0.26 [2.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.13]	0.27 [3.58]	0.26 [3.50]	0.24 [3.20]	0.19 [2.60]	0.17 [2.37]
Quintile	NYSE	EW	0.00 [0.08]	0.03 [0.39]	0.01 [0.11]			
Quintile	Name	VW	0.18 [2.57]	0.23 [3.18]	0.22 [3.14]	0.20 [2.86]	0.16 [2.31]	0.15 [2.09]
Quintile	Cap	VW	0.24 [3.27]	0.27 [3.67]	0.28 [3.72]	0.26 [3.50]	0.21 [2.86]	0.20 [2.70]
Decile	NYSE	VW	0.32 [2.98]	0.38 [3.55]	0.38 [3.60]	0.34 [3.19]	0.26 [2.52]	0.22 [2.20]

Table 3: Conditional sort on size and LFM

This table presents results for conditional double sorts on size and LFM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LFM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LFM and short stocks with low LFM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LFM Quintiles					LFM Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.69 [2.42]	0.95 [3.30]	0.97 [3.39]	1.03 [3.36]	0.86 [3.01]	0.17 [2.35]	0.18 [2.46]	0.16 [2.18]	0.14 [1.84]	0.15 [1.96]	0.13 [1.75]
	(2)	0.78 [2.89]	0.90 [3.48]	0.84 [3.23]	0.93 [3.51]	0.91 [3.59]	0.13 [1.46]	0.18 [2.08]	0.16 [1.88]	0.16 [1.81]	0.15 [1.73]	0.15 [1.71]
	(3)	0.78 [3.18]	0.80 [3.43]	0.84 [3.33]	0.92 [3.84]	0.93 [3.98]	0.15 [1.84]	0.19 [2.31]	0.19 [2.22]	0.15 [1.75]	0.15 [1.72]	0.12 [1.42]
	(4)	0.68 [3.02]	0.90 [4.04]	0.86 [3.69]	0.82 [3.69]	0.85 [3.99]	0.17 [2.22]	0.21 [2.65]	0.18 [2.34]	0.15 [1.93]	0.11 [1.42]	0.10 [1.20]
	(5)	0.46 [2.36]	0.58 [3.11]	0.57 [2.81]	0.69 [3.63]	0.82 [4.29]	0.35 [3.81]	0.37 [3.96]	0.37 [3.92]	0.34 [3.54]	0.28 [2.89]	0.26 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LFM Quintiles					LFM Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	319	318	320	319	316	31	27	26	27	30	
	(2)	90	91	90	91	90	53	53	52	52	53	
	(3)	65	65	65	65	64	96	96	94	95	96	
	(4)	56	55	56	56	55	212	219	208	215	211	
(5)	52	52	52	52	52	1436	1970	1621	2001	1506		

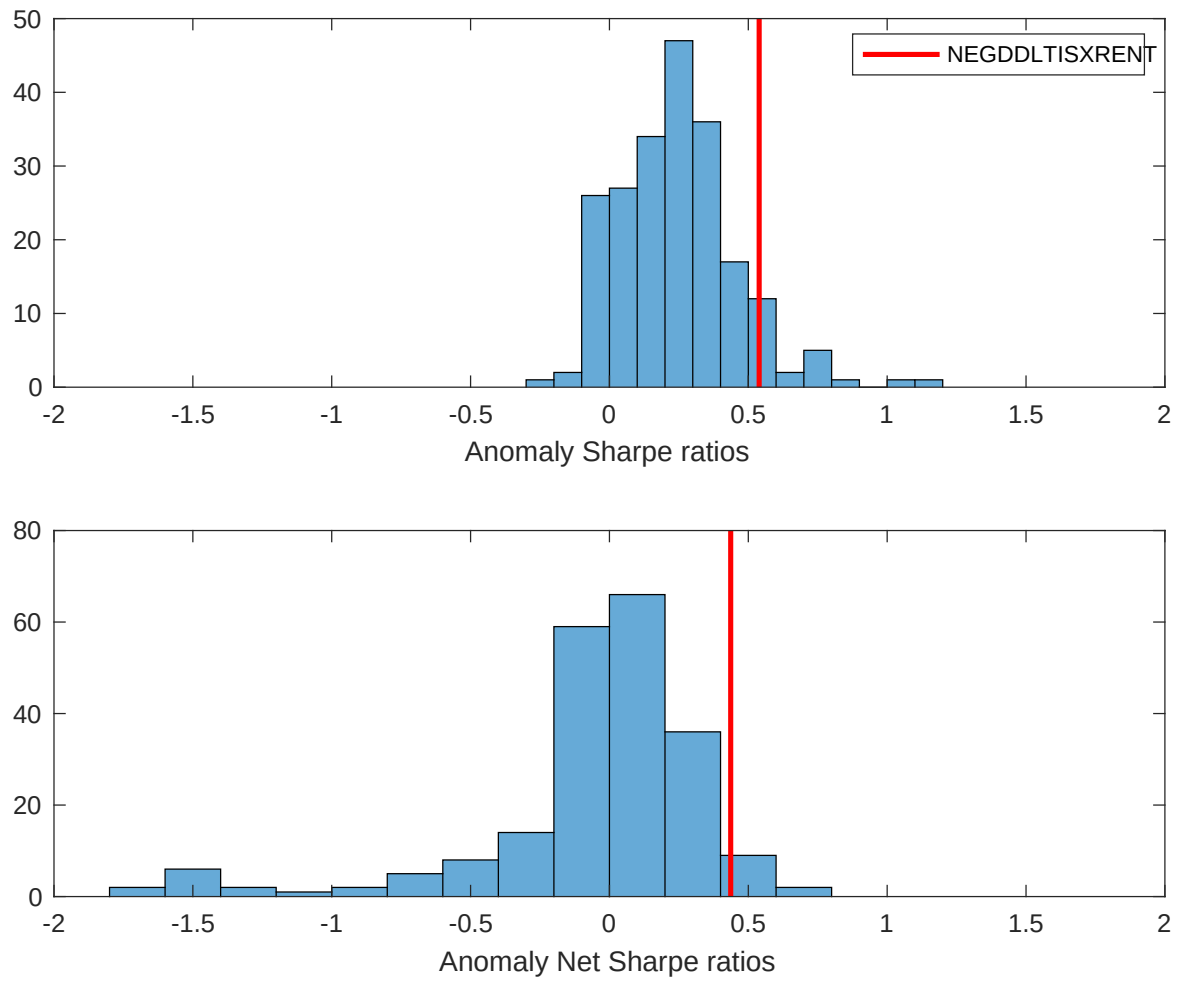


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LFM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

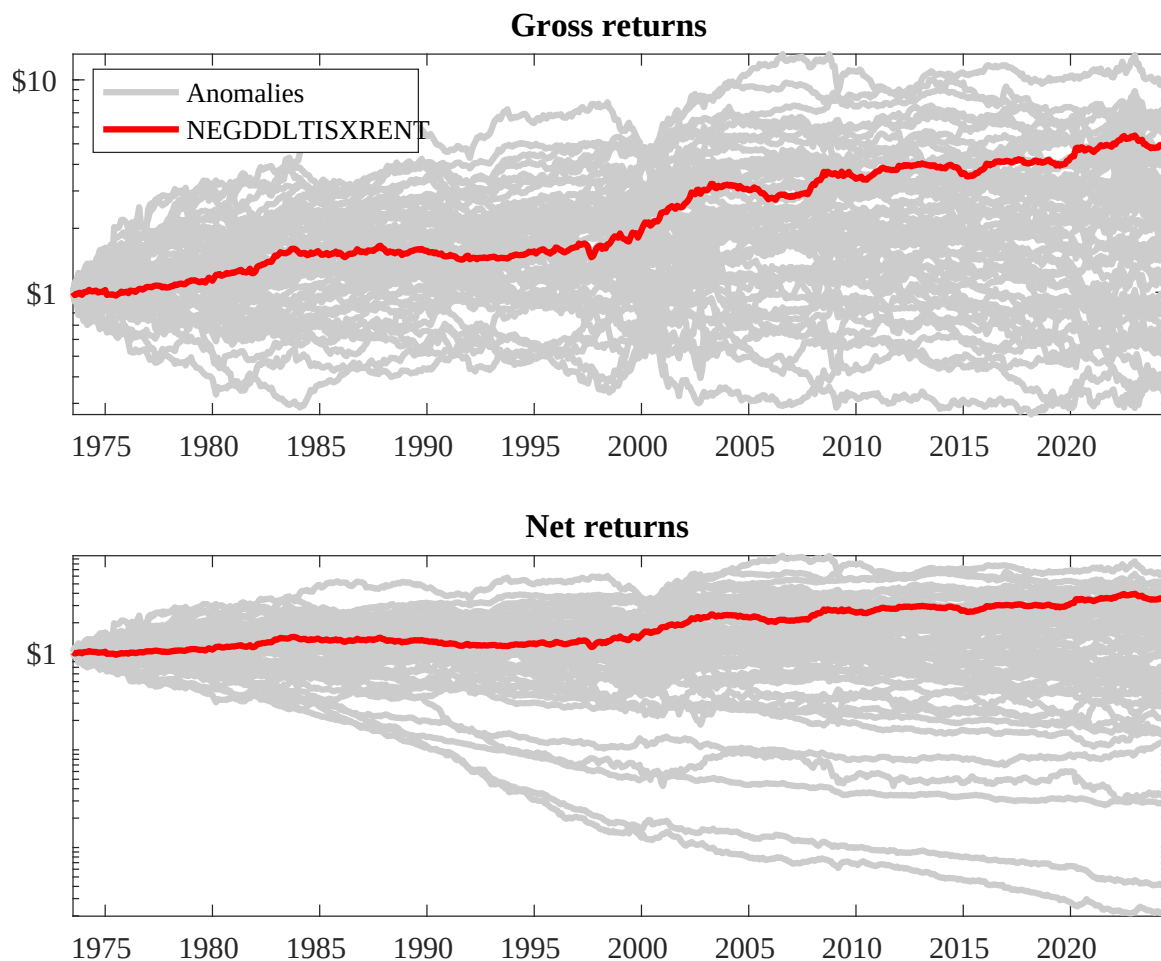
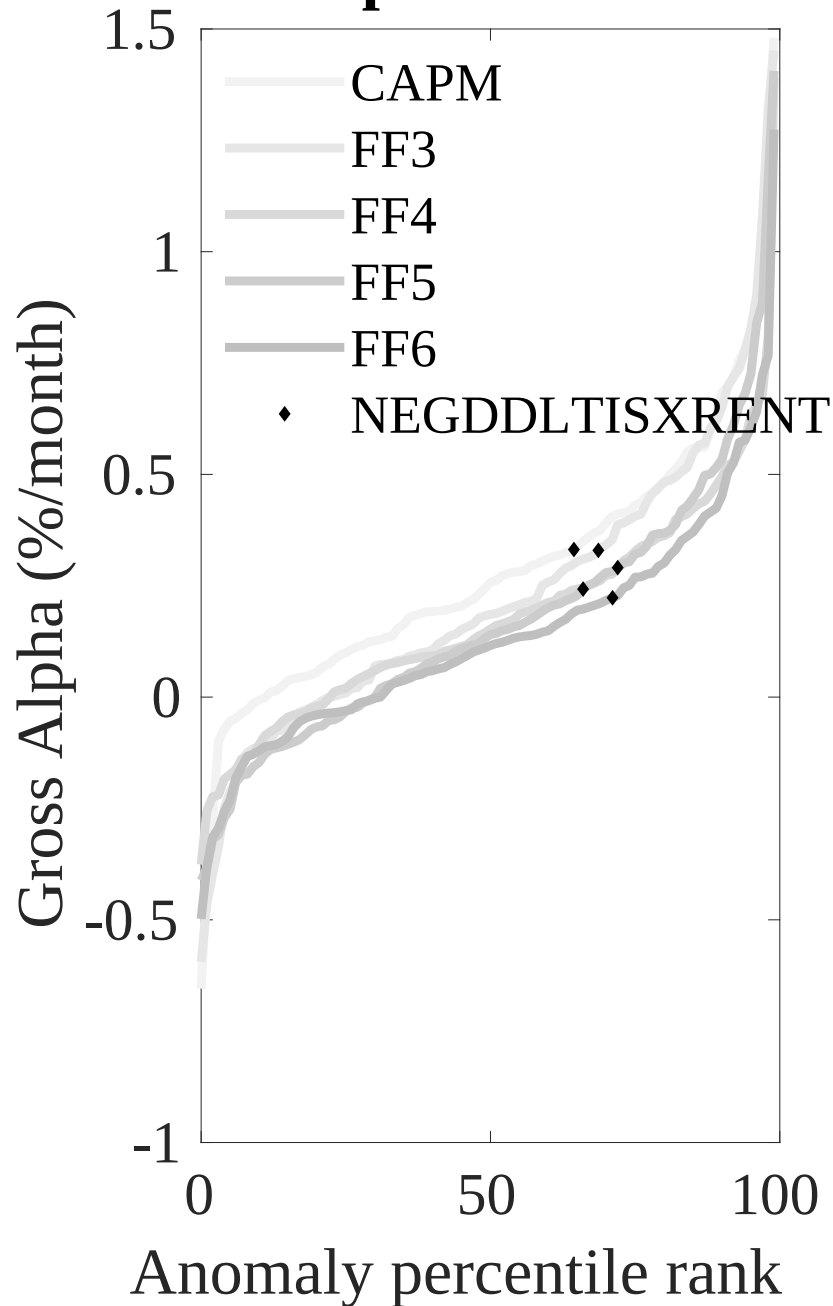


Figure 3: Dollar invested.

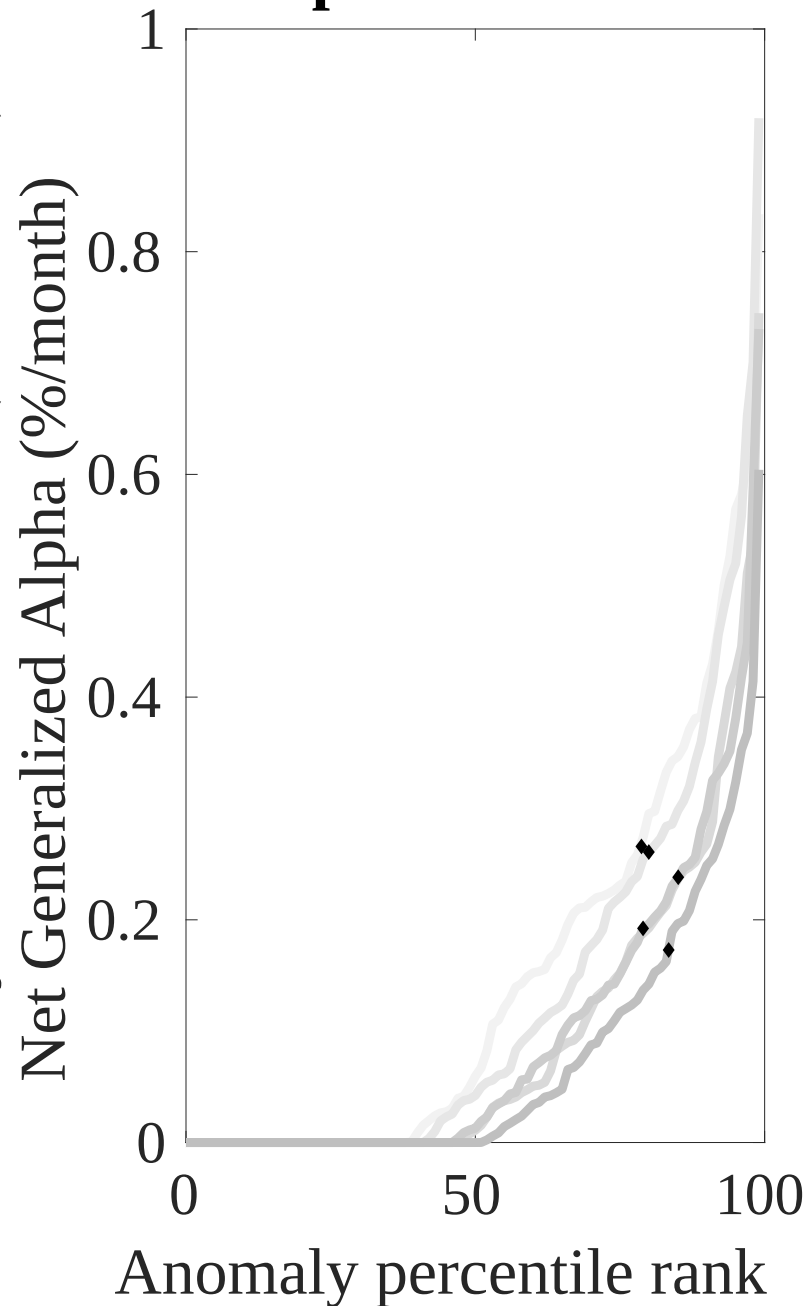
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LFM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



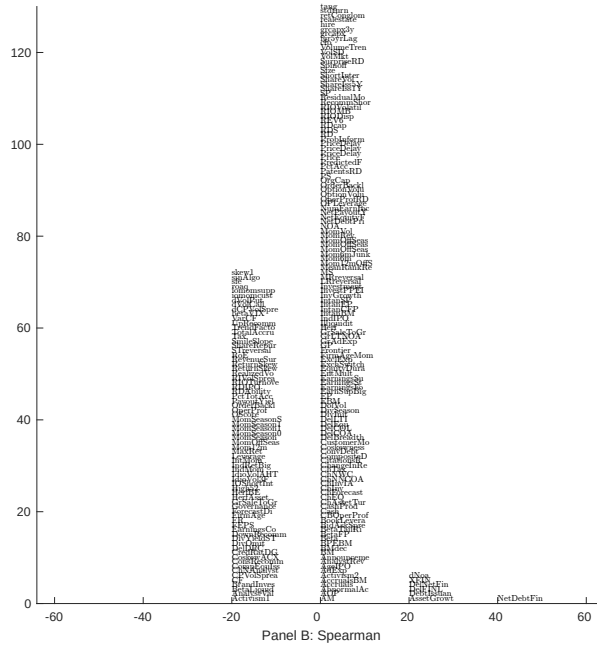
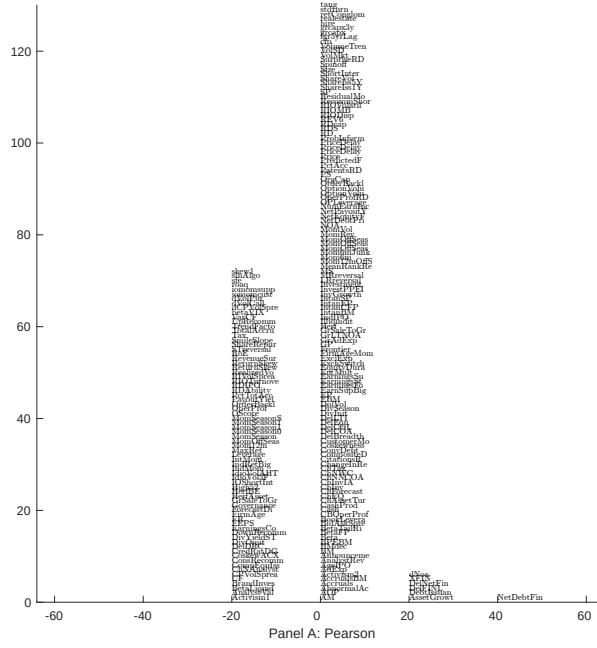


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with LFM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

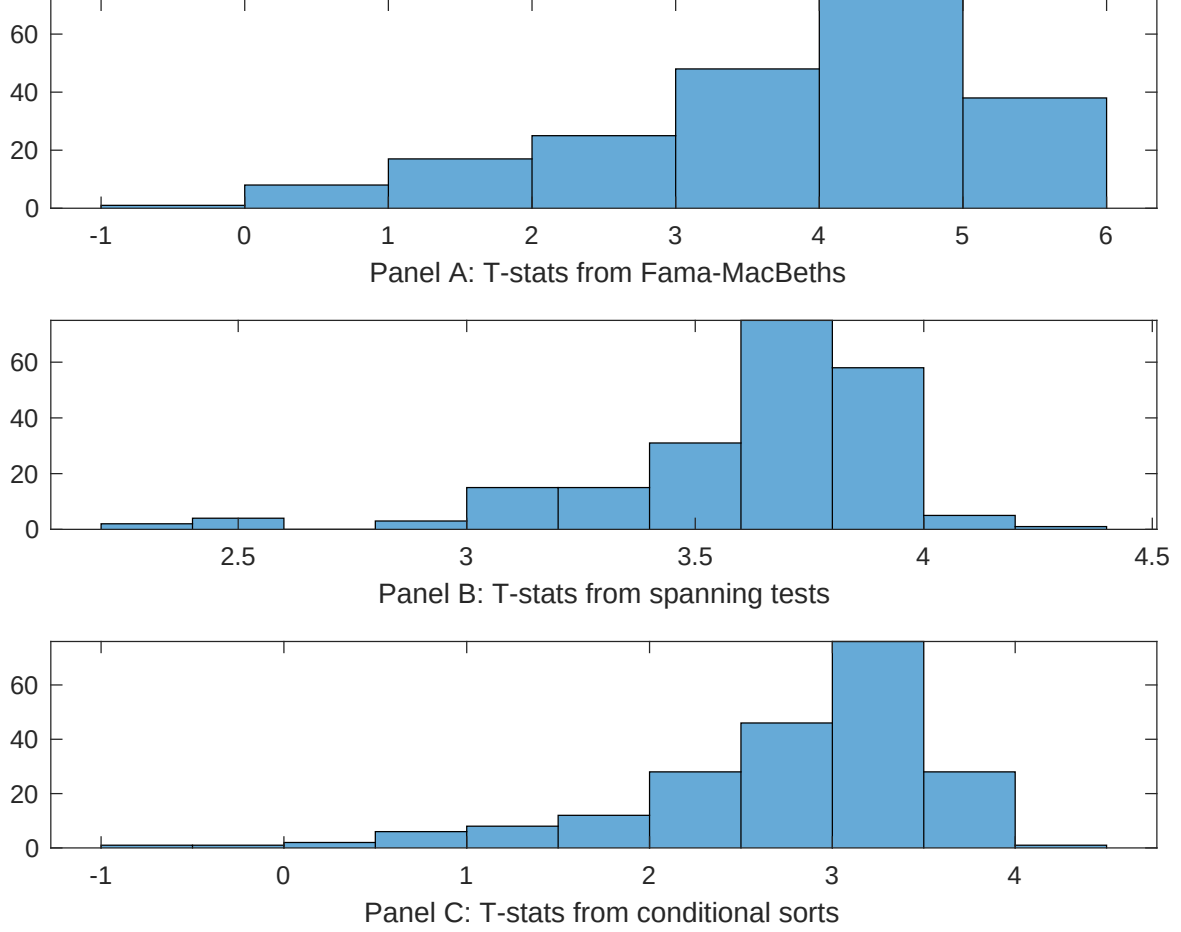


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LFM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [5.34]	0.14 [5.30]	0.15 [5.75]	0.14 [5.69]	0.14 [5.31]	0.14 [5.63]	0.15 [5.71]
LFM	0.35 [0.47]	0.63 [0.85]	0.21 [0.26]	0.12 [1.48]	0.30 [3.19]	0.60 [0.08]	-0.11 [-1.12]
Anomaly 1	0.18 [9.47]						-0.90 [-1.79]
Anomaly 2		0.21 [9.11]					0.84 [1.26]
Anomaly 3			0.11 [8.65]				0.44 [1.90]
Anomaly 4				0.20 [6.58]			0.11 [1.96]
Anomaly 5					0.41 [6.69]		0.17 [0.28]
Anomaly 6						0.15 [9.99]	0.88 [4.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LFM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LFM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.55]	0.19 [2.62]	0.20 [2.73]	0.19 [2.55]	0.21 [2.84]	0.20 [2.65]	0.19 [2.59]
Anomaly 1	15.47 [3.71]						11.40 [1.99]
Anomaly 2		16.01 [3.98]					6.20 [1.13]
Anomaly 3			12.35 [2.88]				7.98 [1.73]
Anomaly 4				14.36 [3.90]			10.23 [2.59]
Anomaly 5					7.79 [2.70]		6.93 [2.36]
Anomaly 6						4.73 [1.11]	-7.08 [-1.49]
mkt	-1.74 [-1.03]	-2.07 [-1.22]	-2.10 [-1.23]	-0.12 [-0.07]	-2.24 [-1.32]	-2.08 [-1.22]	-0.56 [-0.32]
smb	-0.55 [-0.21]	-0.38 [-0.15]	0.13 [0.05]	5.64 [2.00]	1.89 [0.73]	1.12 [0.43]	2.65 [0.89]
hml	-11.74 [-3.59]	-12.11 [-3.71]	-13.35 [-4.06]	-11.17 [-3.40]	-13.01 [-3.97]	-13.16 [-3.97]	-10.65 [-3.23]
rmw	-0.47 [-0.14]	-0.83 [-0.25]	0.97 [0.29]	-7.66 [-1.93]	2.05 [0.61]	1.00 [0.30]	-5.98 [-1.49]
cma	29.37 [5.70]	30.39 [6.01]	19.61 [2.73]	25.18 [4.58]	27.62 [4.94]	31.05 [5.27]	11.90 [1.61]
umd	2.02 [1.18]	2.33 [1.37]	3.62 [2.14]	3.26 [1.94]	2.56 [1.49]	3.21 [1.88]	1.58 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	11	12	11	10	15

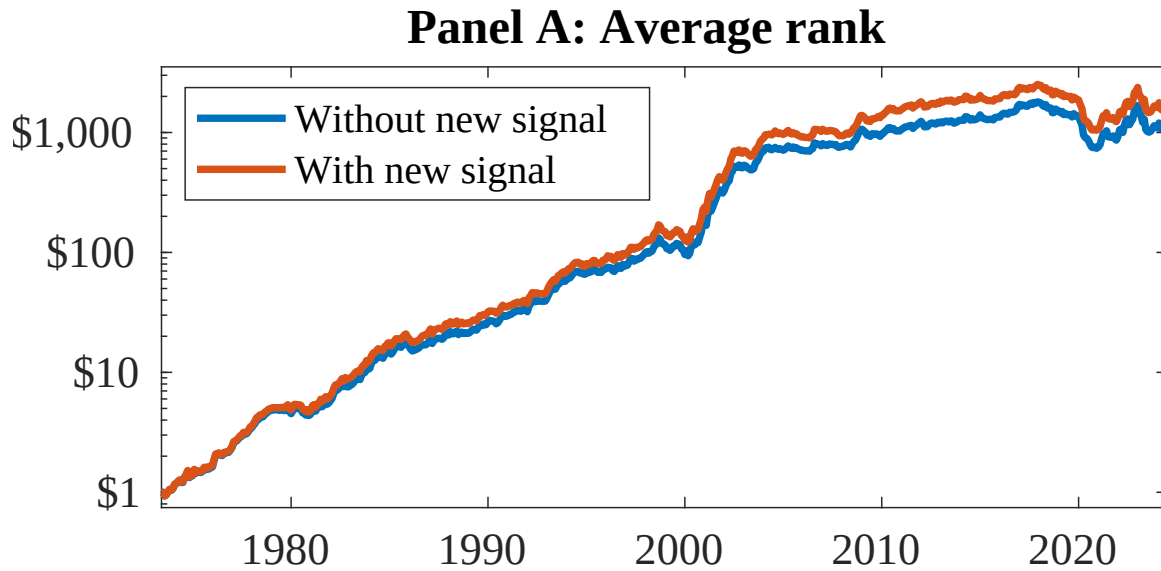


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LFM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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